MIL-T-23369B(AS)
2 January 1968
Superseding
MIL-T-23369A(WEP)
8 March 1965

MILITARY SPECIFICATION

TRANSMITTER GROUP, TELEMETRIC DATA OA-1208( )/DPN

This Specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 SCOPE - This specification covers the requirement for the OA-1208( )/DPN Transmitter Group, Telemetric Data suitable for installation in the AIM-7D, and AIM-7E guided missiles (SPARROW) for use in conjunction with a telemetry system. The OA-1208( )/DPN provides a reliable crystal controlled, single channel, frequency modulated telemetering signal for flight testing of the AIM-7D, and AIM-7E missiles. The electrical variations produced by the missile video, speedgate sweep, and fuzing signals vary the frequency of the radio frequency oscillator, or a subcarrier thereon.

1.2 CLASSIFICATION - The equipment covered by this specification is of one type collectively designated as Transmitter Group, Telemetric Data OA-1208( )/DPN, and consists of the following:

MAJOR UNIT	DESIGNATION	MAX. SIZE	MAX WEIGHT	APPL. PARA.
VHF Transmitter	Unit 1	1.63" x 2.87" x 4.58"	1.1 lb.	3.4.1
Dual Input Voltage Controlled Oscillator and Mixing Circuit	Unit 2	7.0" x 5.0" x 4.0"	1.0 lb.	3.4.2
Antenna Kit	Unit 3		1.0 lb.	3.4.3

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1.3 ACCESSORIES - The equipment covered in this specification is designed to operate without any additional accessories.

2. APPLICABLE DOCUMENTS

2.1 GOVERNMENT FURNISHED DOCUMENTS - The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

Federal Test Method STD NO. 151	Metals; Test Methods

MILITARY

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft; General Specifications for
MIL-E-8189	Electronic Equipment, Guided Missile, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-R-22973	Reliability Index Determination for Avionics Equipment Models; General Specification for

STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electric Components Parts
MIL-STD-210	Climatic Extremes for Military Equipment

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MIL-STD-454 Requirement No. 29	General Requirement for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment; Procedure for Packaging of
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment

BULLETINS

ANA 445 and Supp. No. 1	Engineering Changes to Weapons Systems, Equipment and Facilities

PUBLICATIONS

DEPARTMENT OF DEFENSE	Telemetry Standards Revised March 1966 (IRIG No. 106-66)

DRAWINGS

Oklahoma Aerotronics Inc.

DL 1012134	VHF Transmitter
DL 1012252	Transmitter Group, Telemetric Data OA-1208( )/DPN
DL 1012351	Fuze and Missile Speedgate Sweep Indicator plus EPU Frequency Monitor
DL 1012354	Antenna Kits including One Each of Couplers 5028C-5016 and 5028C-5022

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2.2 AVAILABILITY OF SPECIFICATIONS, PUBLICATIONS AND DRAWINGS -

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Government inspector.

(2) Copies of applicable specifications, and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot (Code 105), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(3) Copies of the Oklahoma Aerotronics, Inc. drawings may be obtained from the Inspector of Naval Material, Torrance, California 90509.

2.3 PRECEDENCE - When the requirements of the contract, this specification, or other applicable specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the Naval Air Systems Command Headquarters.

(3) The drawings listed in this specification.

(4) Referenced Specification - Any specification referenced herein or on the drawings listed in this specification shall have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 PREPRODUCTION REQUIREMENT - Unless otherwise specified in the contract or order, preproduction samples of this equipment shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. No production equipment shall be delivered prior to approval of the preproduction model. Prefabrication of production equipment, prior to such approval, is at the contractor's own risk. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

3.2 TRANSMITTER GROUP - The Transmitter Group, Telemetric Data OA-1208( )/DPN shall be manufactured in accordance with Oklahoma Aerotronics Inc. Drawing DL 1012252 except insofar as errors of omission

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or commission in the technical documentation may be incompatible with the requirements of this specification. In any case a condition of final acceptance is that the transmitter group must meet all the requirements of this specification.

3.2.1 PARTS AND MATERIALS - Since the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will meet material, electrical, and environmental requirements.

3.2.2 CHOICE OF TUBE TYPES - Electronic tubes used in this equipment shall be chosen from Requirement 29 of MIL-STD-454, General Requirement for electronic equipment.

3.2.3 CONFIGURATION - This equipment shall be of sub-miniaturized construction and packaged so as to complement the space, and be compatible with power, mounting, and antenna features provided in the AIM-7D and AIM-7E missiles.

3.2.4 DATA - No data is required by this specification or applicable documents referenced in Section 2 unless specified in the contract or order - see 6.3.

3.3 GENERAL DESIGN REQUIREMENTS

3.3.1 GENERAL - This equipment shall be in accordance with this specification, Specification MIL-E-8189 and the "Telemetry Standards Revised March 1966 (IRIG Document 106-66)."

3.3.2 STANDARD CONDITIONS - The following conditions shall be used for making individual tests on the equipment:

Temperature	Room Ambient 30°C ± 10°C
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Power	As per paragraph 3.3.9

3.3.3 SERVICE CONDITIONS - The equipment shall operate satisfactorily under any of the following service conditions or natural combinations of service conditions.

3.3.3.1 TEMPERATURE - The equipment shall be capable of satisfactory operation when operating at an ambient temperature range from -40°F to +160°F with an initial 10 minute soak period at -40°F. Likewise, the equipment shall operate satisfactorily after a 10 minute soak at +160°F, except that the power output may drop to not less than 0.4 watts when operated at the upper temperature limit.

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3.3.3.2 ALTITUDE - The equipment shall be capable of satisfactory operation from sea level to 90,000 feet of altitude (31.3 to 0.502 inches of mercury), with a soak period for 10 minutes at 90,000 feet.

3.3.3.3 HUMIDITY - The equipment shall be capable of satisfactory operation under relative humidity conditions up to 100%, at temperatures up to 125°F for a soak period of one hour.

3.3.3.4 VIBRATION - The equipment shall operate satisfactorily under vibration applied along longitudinal, vertical, and horizontal axis separately in accordance with MIL-STD-810 Equipment Class 3, Mounting A, Test Curves 3C and 4C, Time Duration of 10 minutes, Omitting Resonance Dwell.

3.3.3.5 SHOCK - The equipment shall be capable of satisfactory operation without damage or maladjustment resulting from impact shock of 50 g's peak acceleration with a load application rate of 35 g's per millisecond and time duration of not less than 5 milliseconds applied along the longitudinal, vertical, and horizontal axis separately.

3.3.3.6 ACCELERATION - The equipment shall be capable of satisfactory operation during and after peak longitudinal acceleration of 40g applied for a minimum of 1 minute; peak longitudinal deceleration of 15g applied for a minimum of 1 minute; and a peak transverse acceleration (any direction normal to that line corresponding to longitudinal axis) of 25g applied for a minimum of 1 minute.

3.3.3.7 COMBINED ENVIRONMENT - The equipment shall operate satisfactorily under any natural combination of conditions specified in the preceding service conditions for a maximum duration of 12 minutes.

3.3.4 FORM FACTOR AND WEIGHT - The form factor of this equipment shall be as follows: The telemetering group shall consist of all units of this equipment, and shall be mounted in the Flight Control Group of AIM-7D or AIM-7E missiles by means of provisions included therein. The weight of the equipment shall not exceed 3.1 pounds.

3.3.5 OPERATIONAL STABILITY -

3.3.5.1 The equipment shall operate under standard conditions in Paragraph 3.3.2 and maintain the characteristics as specified in Paragraphs 3.4.1.4 and 3.4.2.3 for five hours, continuously or intermittently, without necessity for readjustment of any controls. A "warm-up" time shall be allowed in each case prior to use of the system in order that the circuits may reach stable operation. The "warm-up" time shall be as short as possible and shall not exceed a maximum of one-quarter hour.

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3.3.5.2 All tuning and operating controls of the equipment shall be such that when the equipment is properly installed and either operating or non-operating the setting or position of any control shall not be altered by any of the specified service conditions.

3.3.6 OPERATING AND NON-OPERATING LIFE - The equipment shall have a minimum total life of 24 months from date of acceptance by the Navy. The 24 month period may include varying periods of life under the individual conditions of packaged storage, ready storage, inflight readiness, captive flight, and free flight.

3.3.6.1 NON-OPERATING LIFE - The equipment shall have a minimum non-operating life of 24 months from date of Navy acceptance, subject to the provisions of 3.3.6 above. The 24 months may include periods of packaged storage under the conditions of 3.3.3, and periods of unpackaged storage or installed in the missile under depot or shipboard conditions. The equipment shall meet the requirements of MIL-STD-210 with respect to temperature extremes and duration except lower temperature extreme shall be -65°F.

3.3.6.2 OPERATING LIFE - The equipment shall be capable of operating anytime during the 24 month period beginning from the date of Navy acceptance. The 24 month period shall provide a total of at least 50 hours of operation under conditions of 3.3.3.

3.3.7 INTERCONNECTION CABLES - The equipment shall be capable of satisfactory operation using the cables provided as part of the missile and this equipment. The equipment shall operate satisfactorily when used in conjunction with aircraft wiring which has been installed in accordance with Specification MIL-W-5088.

3.3.8 CABLES AND CONNECTORS - The cables and connectors included herein shall facilitate interconnection with the AIM-7D and AIM-7E missile cable in the most efficient manner possible, consistent with the missile design. Special miniature multi-pin connectors may be used to facilitate input and output power connections.

3.3.9 INPUT POWER REQUIREMENTS - The equipment shall operate satisfactorily from the following power sources and shall not require power in excess of the amount indicated:

D.C. Source	Nominal Voltage
B+	+255 VDC at 55 ma
+25 VDC filament	+25 VDC at 600 ma

3.3.9.1 D.C. VOLTAGE LIMITS - The voltage limits for B+ shall be 255 ± 5 VDC; however, the equipment shall not be damaged by voltages

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as high as 285 VDC. The equipment shall operate satisfactorily with a filament supply of 24 to 26 VDC.

3.3.9.2 GROUND INPUT - The equipment shall be such that the negative of the primary DC power source may be grounded externally.

3.3.10 OVERLOAD PROTECTION - Overload protection for the equipment need not be provided in the equipment.

3.3.11 UNDER VOLTAGE PROTECTION - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within the limits of 3.3.9 above.

3.3.12 MOISTURE AND FUNGUS RESISTANT TREATMENT - Moisture and fungus resistant treatment is not required for this equipment.

3.3.13 NAMEPLATES AND NOMENCLATURE - Application for nomenclature and marking of equipment for identification shall be in accordance with Specification MIL-N-18307.

3.3.14 INTERCHANGEABILITY - The equipment shall be designed to meet the interchangeability requirements of Specification MIL-E-8189 to the subassembly level.

3.3.15 PROVISIONS FOR MAINTENANCE AND FIELD TESTING - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-E-8189 as far as possible, consistent with the nature of this equipment. No field maintenance shall be performed on the VHF transmitter, Unit 1 or the 3.9 Kc modulator, Unit 2.

3.3.16 INTERFERENCE - Every effort shall be made, consistent with its size and function, to minimize unwanted radiation and susceptibility to interference within the limits specified in MIL-STD-461.

However, in any case, the antenna conducted and radiated interference shall conform with specifications set forth in IRIG Document 106-66 when operated in any service or load conditions specified herein.

3.3.17 PERFORMANCE - The equipment to be furnished under this specification shall exhibit radio frequency modulation characteristics as follows: The VHF transmitter of this equipment shall be directly phase or frequency modulated by wide band video, 3.9 Kc modulator and 1200 cycles information as extracted from the missile EPU. This equipment shall include provisions for coupling the frequency moculator carrier output to an antenna for transmission to a receiver-recorder system.

3.3.17.1 - FREQUENCY - The transmitting equipment shall be capable of being factory adjusted to permit operation on a crystal controlled carrier frequency in the range of 225 to 260 megacycles.

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3.3.17.2 OUTPUT - Except as modified in 3.3.3.1, the radio frequency output shall be no less than 0.5 watts.

3.3.17.3 MODULATION - Wide band phase or frequency modulation techniques shall be employed; the overall frequency excursion of the radio frequency carrier shall not exceed ± 100 kilocycles per second as a result of the video or steady state fuzing subcarrier modulation. Intermodulation and harmonic distortion products shall be at least 34 db down from the video signal when operated under the service conditions outlined in 3.3.3.

3.3.17.4 CARRIER DRIFT - Frequency change (after warm-up) due to drift shall not exceed ± 0.01% of the mean center frequency.

3.3.18 WORKMANSHIP - The workmanship shall conform to Specification MIL-E-8189. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by the applicable specification.

3.4 DETAIL DESIGN REQUIREMENTS

3.4.1 VHF TRANSMITTER (UNIT 1) - The VHF transmitter, Oklahoma Aerotronic, Inc. Drawing DL 1012134, shall meet the following requirements:

3.4.1.1 FUNCTION - This unit shall be a VHF transmitter operating in the range of 225 to 260 megacycles with an output power NLT 0.5 watts into a matched load, except as modified in 3.3.3.1. Phase or frequency modulation characteristics shall be employed to continuously transmit missile video, speedgate sweep, EPU, and fuze data to a receiver-recorder system.

3.4.1.2 FORM FACTOR AND MOUNTING - This unit shall not be greater than 1.63 inches by 2.87 inches by 4.58 inches and shall be capable of installation in the Flight Control Group of the AIM-7D and AIM-7E missile. The mounting brackets shall be provided as a part of this unit.

3.4.1.3 CONTENTS - This unit shall contain a modulator circuit, crystal controlled oscillator, frequency multipliers, and amplifiers.

3.4.1.4 CHARACTERISTICS - This unit shall have the following nominal characteristics:

(a) Power output	No less than 0.5 watts.
(b) Frequency	225 mc to 260 mc.
(c) Modulation response	1 Kc to 150 Kc minimum. If phase modulation is utilized, the frequency response shall be compensated to provide flat (± 1 db) response above 20 Kc.

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(d) Modulation sensitivity	This shall be consistent with the requirements of Section 70.0 of Appendix I.
(e) RF output impedance	50 ohms nominal.
(f) Power input	Both B+ and filament voltages shall be adjustable by dropping resistors.
(g) Spurious Radiation	Shall comply with the requirements of 3.3.16 when operated into an antenna system having a standing wave ratio (SWR) of 1.5 to 1, (inductive, resistive or capacitive) referenced to 50 ohms.

3.4.1.5 ELECTRICAL CONNECTIONS - This unit shall have the following electrical connections:

(a) RF output UG-625/U (J2) mates with UG-88/U

(b) Input (Winchester plug M5P) or an approved equivalent

3.4.1.6 WEIGHT - The weight of this unit shall not exceed 1.1 pounds.

3.4.2 DUAL INPUT VOLTAGE CONTROLLED OSCILLATOR AND MIXING CIRCUIT (UNIT 2) - This unit, Oklahoma Aerotronics Inc. Drawing DL 1012351, shall meet the following requirements:

3.4.2.1 FUNCTION - This unit shall modulate the transmitter with the mixed output of the VCO, video doppler, and EPU frequency.

3.4.2.2 FORM FACTOR AND MOUNTING - This unit shall not be greater than 7.0 inches by 5.0 inches by 4.0 inches (including mounting bracket).

3.4.2.3 CHARACTERISTICS - This unit shall have the following characteristics after a 15 minute warm-up:

3.4.2.3.1 VCO CHARACTERISTICS

(a) Center frequency	3900 $\pm$ 8 cps (with 2.5 VDC input).
(b) Deviation	$\pm$ 7.5% nominal, for 0 to 5 VDC input.
(c) Linearity	Maximum 2% best straight line.
(d) Input #1	Direct coupled with minimum input impedance of 500 K ohms.
(e) Input #2	AC coupled pulse stretcher with a 20 millisecond duration.
(f) Input isolation	Isolation between input #1 and input #2 must be minimum of 200 K ohms.
(g) Output impedance	Compatible with mixing circuit input.

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3.4.2.3.2 MIXING CIRCUIT CHARACTERISTICS

(a) Input impedance.	Minimum 50 K ohms (video input must have switchable mixing levels. See drawing DL 1012351).
(b) Output impedance	Compatible with transmitter input.
(c) Isolation between inputs	Minimum 200 K ohms.
(d) Intermodulation	All intermodulation must be 34 db below level of highest input signal.

3.4.2.4 OSCILLATOR DRIFT - After a 15 minute warm-up the frequency shall be adjusted to within ± 8 cps of the 3.9 Kc center frequency. The frequency of this oscillator shall not drift more than ± 8 cps of the initially set frequency for a period of five (5) hours under the standard conditions of paragraph 3.3.2; nor change more than ± 18 cps when subjected to the service conditions of paragraph 3.3.3.

3.4.2.5 ELECTRICAL CONNECTIONS - This unit shall have the following electrical connections:

(a) Input, Amphenol 165-15 or an approved equivalent.

(b) Safety and Arming Connector, Amphenol 165-16 or an approved equivalent.

(c) Missile Speedgate Sweep, Winchester SA15S or an approved equivalent.

3.4.2.6 WEIGHT - The weight of this unit including the bracket shall not exceed 1.0 pounds.

3.4.3 ANTENNA KIT (UNIT 3) - This unit, Oklahoma Aerotronics Inc. Drawing DL 1012354, shall meet the following requirements:

3.4.3.1 FUNCTION - This unit shall include the antenna, the antenna couplers and necessary attaching parts for same, a means of impedance matching the antenna with the transmitter, and the coaxial cables for connecting the transmitter to the antenna couplers. In addition it is required that a table of optimum antenna lengths for the operating frequencies, such that a VSWR of 1.5 or better can be obtained from the antenna-coupler combination, shall be supplied.

3.4.3.2 WEIGHT - This unit shall not exceed 1.0 pound.

3.5 RELIABILITY IN MEAN TIME BETWEEN FAILURES (MTBF) - For the test program under 4 the mean (operating) time between failures shall be 50 hours.

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4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF INSPECTIONS - Inspection shall be classified as follows:

(a) Preproduction inspection.

(b) Quality conformance inspection.

(c) Periodic inspection.

4.3 SAMPLING

4.3.1 PREPRODUCTION SAMPLE - A preproduction sample of five equipments, manufactured in accordance with the requirements of section 3, shall be subjected to the preproduction inspection detailed in 4.4 at a laboratory specified by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the equipment, by the supplier, prior to approval of the preproduction sample shall be at the suppliers risk.

4.4 PREPRODUCTION INSPECTION - The preproduction sample, after satisfactorily passing the operational test, examinations, and manufacturing run-in test of 4.5 shall be subjected to the following:

(a) Environmental Tests.

(b) Radio Interference Test.

(c) Reliability Tests.

4.4.1 ENVIRONMENTAL TESTS - Environmental tests, to verify the requirements of 3.3, shall be as detailed in MIL-STD-202, as modified by this specification.

4.4.2 RADIO INTERFERENCE TESTS - Radio interference tests shall be as detailed in Specification MIL-STD-461 .

4.4.3 RELIABILITY TESTS - Reliability tests shall be as detailed in Specification MIL-R-22973 with the following additions and exceptions.

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4.4.3.1 QUANTITY OF EQUIPMENT TO BE TESTED - Five preproduction prototypes of the equipment shall be tested.

4.4.3.2 LENGTH OF TESTS - The total hours of test, not including any burn-in period, shall be 50 hours.

4.4.3.3 TEST LEVEL - Test level III as listed in Table I of MIL-R-22973 shall apply except as follows:

4.4.3.4 DUTY CYCLE - Shall be as shown:

Test Conditions

-40°F at 30,000 feet.
-40°F at 90,000 feet.
+160°F at 30,000 feet.
+160°F at 90,000 feet.

Each test shall consist of 10 minutes of non-operating temperature soak at each specified temperature and altitude followed by 5 minutes of operating time.

4.4.3.5 PERFORMANCE CHARACTERISTICS - The equipment shall comply with the performance characteristics of paragraphs 3.3 and 3.4 of this specification.

4.4.3.6 EQUIPMENT FAILURE - The criteria for equipment failure shall be as defined in paragraph 3.12 of MIL-R-22973.

4.4.3.7 MAINTENANCE - Maintenance shall be limited to that specified in paragraph 3.9 of MIL-R-22973. Preventative maintenance shall not be performed.

4.4.3.8 REPLACEMENT PARTS - The replacement of parts shall be governed by paragraph 3.10 of MIL-R-22973.

4.4.3.9 DISPOSITION OF EQUIPMENT - The disposition of equipment shall be as specified by the procuring activity.

4.5 QUALITY CONFORMANCE INSPECTION - Quality conformance inspection shall consist of the following:

(a) Examination.

(b) Manufacturing Run-in Test.

(c) Operational Test.

(d) Reliability Assurance Tests.

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Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 EXAMINATION - Each equipment shall be examined to verify compliance with the requirements for material, workmanship, form factor, weight, content, and presence of applicable controls, displays and connections set forth in this specification.

4.5.2 OPERATIONAL TEST - Each equipment shall be operated long enough to permit the temperature to stabilize, and subjected to the tests set forth in Appendix I. Failure to meet any test requirement shall be cause for rejection of the equipment.

4.5.3 MANUFACTURING RUN-IN TEST - Each equipment shall be operated under the conditions specified herein for a period of 1 hour without failure. A failure shall be defined as anything which causes malfunction of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during normal use of the equipment.

Temperature	Ambient room.
Humidity	Ambient room.
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of $\pm$ 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 3 minutes prior to the beginning of the 1 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. Direction of vibration should be vertical to the normal mounting plane. Should a failure occur, it should be repaired and the test started over, except that the 3 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures (see 6.3).

4.5.4 RELIABILITY ASSURANCE TESTS - Reliability assurance tests shall be conducted on production equipments which have successfully passed the inspection of 4.5.1, 4.5.2, and 4.5.3.

4.5.4.1 SCOPE OF TESTS - Reliability assurance tests shall be conducted in accordance with Specification MIL-STD-781 Test Level A, unless otherwise specified.

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4.5.4.2 ACCEPT-REJECT CRITERIA - Shall be as specified in paragraph 4.2.8.1 of MIL-STD-781.

4.5.4.3 DUTY CYCLE - Shall be as shown (see 4.4.3.4).

4.5.4.4 PERFORMANCE CHARACTERISTICS - Each equipment subjected to the reliability assurance tests shall meet the performance requirement of Appendix I.

4.5.4.5 EQUIPMENT FAILURE - The criteria for equipment failure shall be as defined in paragraph 5.5 of MIL-STD-781.

4.5.4.6 CORRECTIVE ACTION - Corrective action and/or preventative maintenance shall be limited to that specified in paragraphs 5.8 and 5.9 of MIL-STD-781A.

4.5.4.7 RESTORATION - Restoration of failed equipments shall be governed by paragraph 5.9 of MIL-STD-781.

4.5.4.8 DISPOSTION OF EQUIPMENT - The disposition of equipment shall be as specified by the procuring activity.

4.6 PERIODIC INSPECTION - The periodic production samples selected at random shall be subjected to the environmental, radio interference, examination, manufacturing run-in, and operational tests specified in 4.4.1, 4.4.2, 4.5.1, 4.5.2, and 4.5.3. Acceptance shall be based on equipment meeting all the requirements specified.

4.6.1 FAILURE REPORT - In the event of a failure, the Government representative shall be notified immediately. When specified in the contract or order (see 6.3), a report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity shall review such proposals and determine whether they are acceptable.

5. PREPARATION FOR DELIVERY

5.1 GENERAL - All major units and parts of the equipment shall be preserved, packaged, packed and marked in accordance with Specification MIL-E-17555. The level(s) of packaging and packing required shall be as specified in the contract. In the event any of the equipment is not listed in Specification MIL-E-17555, the method of preservation for level A shall be determined by use of the section chart included in Appendix D to MIL-STD-794.

6. NOTES

6.1 USE - The equipment specified herein is for use in the AIM-7D and AIM-7E guided missile to transmit missile flight data to telemetry receiving stations.

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6.2 PERFORMANCE OBJECTIVES - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 ORDERING DATA - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of application level of packaging and packing required (see paragraph 5.1).

(c) Data required for operational use and for maintenance and procurement purposes.

(d) Select which quality assurance tests are to be performed:

(1) Preproduction
(2) Manufacturing run-in
(3) Periodic inspection
(4) Reliability assurance.

(e) Designate disposition of equipment used under Section 4 of this specification.

6.3.1 DESIGN DATA - The contractor should furnish only those items of the following data that are specified in the contract or order.

6.3.2 QUALITY CONFORMANCE INSPECTION DATA - The contractor shall furnish test reports showing quantitative results of quality conformance tests (paragraph 4.5). Such reports should be signed by an authorized representative of the contractor or laboratory, as applicable.

6.3.2.1 SPECIAL TEST PROCEDURES - Where previously approved test procedures for Production and Sampling tests are not applicable to required Special Tests, the contractor should prepare a test procedure and submit it to the procuring activity for approval prior to conducting such tests.

6.3.2.2 EQUIPMENT FAILURE - Should a failure occur during sampling tests, the contractor shall submit to the procuring activity, proposed corrective

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action intended to reduce the possibility of the same failure(s) occurring in future tests (see paragraph 4.3.2).

6.3.3 LIFE TEST DATA - The life test failure record is specified in 4.6.1.4.

6.3.4 REJECTION AND RETEST - Before resubmitting equipment which was rejected and subsequently reworked, full particulars concerning previous rejection and the action taken to correct the defects found in the original should be furnished the government inspector.

6.3.4.1 RECONDITIONING OF LIFE TEST SAMPLES - When life testing is required, an equipment which has been subjected to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

6.3.5 NOMENCLATURE, NAMEPLATES, AND SERIAL NUMBERS - The contractor shall request assignment of nomenclature-type designations, serial number, and nameplate approval to identify the equipment and major units to be furnished under the contract in accordance with Specification MIL-N-18307.

6.3.6 PREPRODUCTION TEST REPORTS - Following the Preproduction Tests, three copies of a report presenting data obtained by the contractor in conducting the Preproduction Tests (paragraph 4.4) should be submitted to the procuring activity for review and approval. Included in this report should be an analysis of all failures which occurred, and suggestions for improvements in design which might be incorporated in later productions. The Preproduction Test Reports shall report on all of the appropriate tests referenced under paragraph 4.4 and following subparagraphs.

6.3.7 EQUIPMENT CHANGES - After approval of the preproduction model, any changes which affect weight, installation interchangeability, and interchangeability of maintenance parts of assemblies shall require prior approval of the procuring activity. To obtain procuring activity approval, the contractor should submit an Engineering Change Proposal giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. Engineering Change Proposals should be prepared in accordance with ANA Bulletin 445 including Supplement No. 1.

6.3.8 INSTALLATION INSTRUCTIONS - One copy of installation instructions shall be provided with each equipment.

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6.4 NOMENCLATURE - The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.3.13. The complete type number shall be used on nameplates, shipping records, and instruction books as applicable.

6.5 WARNING TO CONTRACTORS - Where the requirements of this and related specifications refer to information to be given in the contract, or to information to be furnished by the procuring activity on request, and if such information is not given in the contract, it shall be the responsibility of the bidders and contractors to request this information by official correspondence. Such information shall not be binding on the procuring activity unless it is forwarded to the contractor by official correspondence.

6.6 CONFLICTING REQUIREMENTS - Conflicting requirements arising between this specification and any specifications, publications, or drawings listed herein shall be referred in writing to the procuring activity or its appointed agent for interpretation and clarification.

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APPENDIX I

TRANSMITTER GROUP, TELEMETRIC DATA OA-1208( )/DPN

MANUFACTURING TEST PROCEDURE

10.0 PRELIMINARY

10.1 Visually verify that no damage has occurred during shipment. (1)_____ Initial

10.2 Verify that the unit bears proper mechanical inspection tags. (2)_____ Initial

10.3 Record unit serial number on Test Data Sheet. (3)_____ Initial

10.4 Record the transmitter type number on the Test Data Sheet. (4)_____ Initial

10.5 Verify that the Test Data Sheet information has been properly entered. (5)_____ Initial

20.0 TEST EQUIPMENT

20.1 25 VDC ± 1% at 5A regulated power supply.

20.2 240 to 280 VDC ± 2% at 0.1A regulated power supply.

20.3 2.5 VDC and 5.0 VDC ± 1% regulated power supply.

20.4 0-100 milliameter.

20.5 Frequency Counter, Beckman Model 5230R.

20.6 AC VTVM, Ballantine 310A.

20.7 DC VTVM, Simpson 303.

20.8 2 Wide Range Oscillators, Hewlett Packard 200CD.

20.9 Receiver, Nems-Clarke 1670-E.

20.10 Oscilloscope, Dumont 401-R.

20.11 RF Wattmeter, Thruline Model 43.

20.12 Load Box, Bendix Special Assembly No. 3145821.

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20.13 Telemetering Indicator, Panoramic Model TMI-1B.

20.14 FM-AM Signal Generator, Boonton Type 202J, 195-270 mc.

20.15 RF Counter, Beckman Model 6146 with Model 607 plug in.

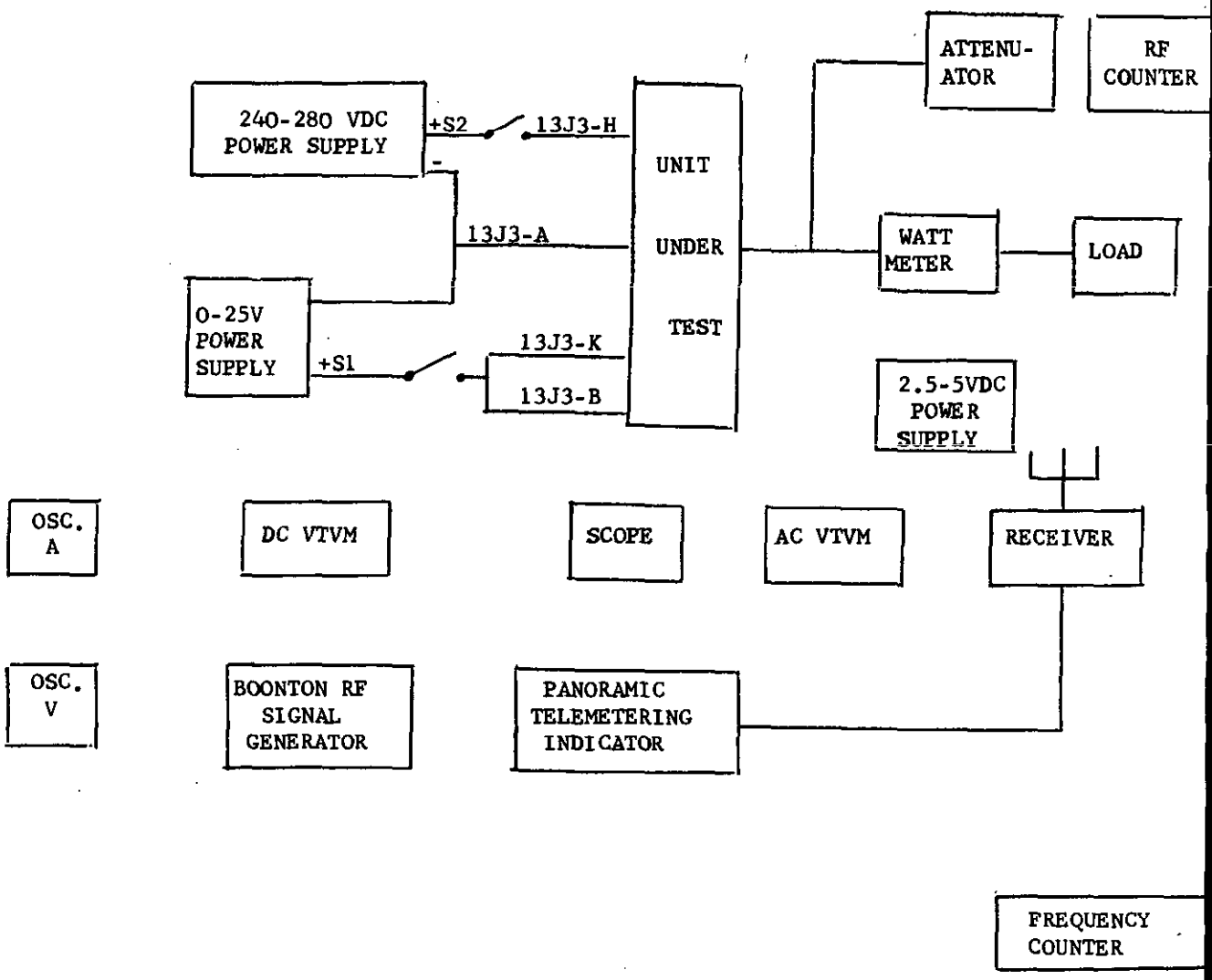
20.16 RF Attenuator.

NOTE: Test equipment of equivalent or better accuracy may be substituted for the above listed test equipment.

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30.0 EQUIPMENT SET-UP

30.1 Connect the test equipment according to the following diagram.



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30.2 Connect the test equipment to the appropriate power source. Turn the test equipment ON and allow a thirty minute warm-up period.

30.3 Disconnect the 13 P1 plug from the 13 J1 jack on the OA-1208( )/DPN transmitter.

30.4 Connect the DC VTVM across the 25 VDC power supply and turn S1 ON. Adjust the power supply for +25 volts as indicated on the DC VTVM.

30.5 Remove the leads of the DC VTVM from the 25 VDC power supply and reconnect them across the 240-280 VDC power supply.

30.6 Wait 30 seconds, then turn S2 ON. Adjust 240-280 VDC power supply for +260 volts as indicated on the DC VTVM.

30.6.1 Maintain the DC voltage source at +260 VDC at all times.

30.6.2 Whenever it is necessary to change a nominal resistor, remove the power from the unit by first placing the B+ switch (S2) to OFF.

30.7 Turn S1 and S2 OFF.

40.0 MODULATOR FREQUENCY AND DEVIATION

40.1 Install nominal resistors R114, R115, R118, R132 and R136.

SYMBOL	FUNCTION	NOMINAL
R114	Fine adjustment for frequency deviation	2 megohm
R115	Coarse adjustment for frequency deviation	309K ohm
R118	Center frequency adjustment	130K ohm
R132	Adjusts modulator input level to transmitter and affects transmitter deviation	3.9K ohm
R136	Adjusts transmitter deviation level	2.1K ohm

40.2 Turn S1 to ON.

40.3 Wait 30 seconds, then turn S2 ON.

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40.4 Connect the output of the 2.5-5.0 VDC power supply between 1OP1B pin E and chassis ground of the OA-1208( )/DPN.

40.5 Connect the DC VTVM between 1OP1B pin E and chassis ground of the OA-1208( )/DPN.

40.6 Adjust the 2.5-5.0 VDC power supply for 2.5 ± .25 VDC as observed on the DC VTVM.

40.7 Connect the frequency counter input leads between the junction of R132-C126 and chassis ground of the OA-1208( )/DPN modulator assembly.

40.8 Adjust the potentiometer R117 for a center frequency of 3900 ± 8 cps indication on the frequency counter.

(6a) _____
2.5V ± 1%

(6b) _____
3900 ± 8 cps

40.9 Adjust the 2.5-5.0 VDC power supply for 5 volts ± 1% indicated on the DC VTVM. The frequency counter should now indicate 4185 ± 8 cps.

(7a) _____
5.0V ± 1%

(7b) _____
4185 ± 8 cps

40.10 Disconnect the 2.5-5.0 VDC power supply lead from 1OP1B pin E, and use a clip lead to connect 1OP1B pin E to chassis ground of the OA-1208( )/DPN. The DC VTVM should read 0.0 volts and the frequency counter should indicate 3616 ± 8 cps.

(8a) _____
0.0 VDC

(8b) _____
3616 ± 8 cps

40.11 Adjust resistors R114, R115 and R118 to obtain optimum results in steps 40.8, 40.9, and 40.10.

40.12 Disconnect the frequency counter from the OA-1208( )/DPN modulator assembly.

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40.13 Set the AC VTVM to the 1-volt range, and connect its input leads between the junction of R132-R131 and chassis ground of the OA-1208( )/DPN.

40.14 Adjust R132 to obtain a 0.150 ± .02 VAC indication on the AC VTVM.

(9) _____ VAC
0.15 ± 0.02 VAC

40.15 Remove clip lead from 1OP1B pin E and chassis ground.

40.16 Reconnect the 2.5-5.0 VDC power supply between 1OP1B pin E and chassis ground on the modulator and adjust it for 5.0 ± 0.05 VDC as indicated on the DC VTVM.

40.17 The reading on the AC VTVM should not drop more than 3db from the reading in 40.14.

40.18 Disconnect the DC VTVM, the 2.5-5.0 VDC power supply and the AC VTVM from the modulator assembly.

50.0 FUZE

50.1 Connect the oscilloscope's input leads between the junction of R132-C126 and chassis ground on the OA-1208( )/DPN modulator assembly.

50.1.1 Verify that the CRT display is a sine wave free from limiting.

(10) _____
Initial

50.2 Insert a 25 volt spike into the OA-1208( )/DPN fuze circuit (13J3 pin D) and verify that the CRT display indicates this fuze spike.

(11) _____
Initial

50.2.1 The fuze spike may be obtained from the circuit shown in Figure 1.

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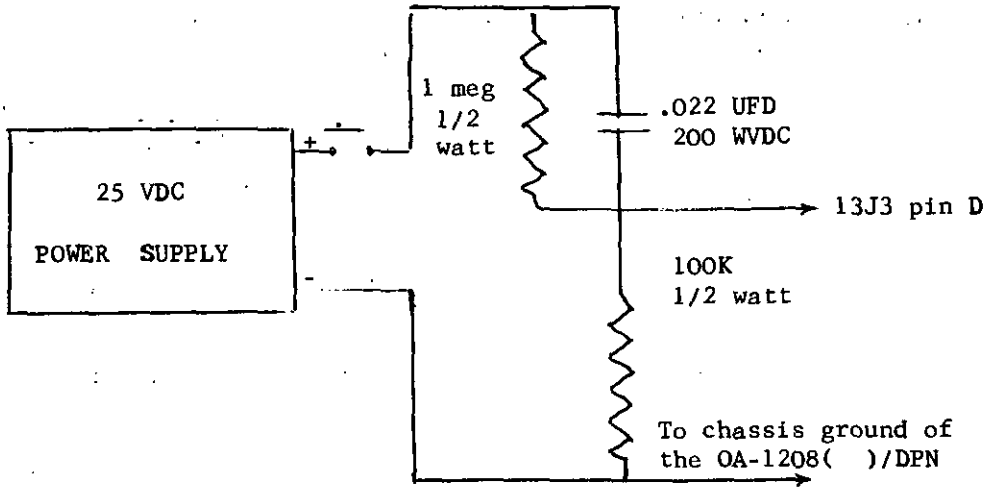


Figure 1

70.0 TRANSMITTER DEVIATION

70.1 Remove the DC VTVM's input leads from the OA-1208( )/DPN and reconnect them between 10P1B pin E and chassis ground.

70.2 Connect the 2.5-5.0 VDC power supply between 10P1B pin E and chassis ground. Adjust the 2.5-5.0 VDC power supply for +2.5 volts as observed on the DC VTVM.

70.3 Connect the AC VTVM's input leads between the junction of R130-K101 pin 3 and chassis ground.

70.4 Set the output frequency of "Oscillator A" to 40 Kc and connect its output leads between 13J3 pin E and chassis ground. Adjust its output voltage to 475 millivolts AC as observed on the AC VTVM.

(15) _____
Initial

70.5 Disconnect the AC VTVM from junction R130-K101-3 and reconnect it to junction R134-13J3 pin J and chassis ground.

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70.6 Set the output frequency of "Oscillator B" to 1200 cps and connect its output leads between 13J3 pin J and chassis ground. Adjust its output voltage for 9 volts AC as indicated on the AC VTVM.

(16) _____
Initial

70.7 Tune the receiver to the transmitter operating frequency. Frequency must be in the range of 225 to 260 megacycles.

(17) _____
Initial

70.8 Turn S2 off.

70.9 Connect the output of Boonton Signal Generator to the receiver input.

70.10 Make the following adjustments on the Boonton Signal Generator.

70.10.1 Set "DBM-Microvolt Control" to 5K.

70.10.2 Tune the Boonton to the transmitter operating frequency as indicated on the receiver.

70.10.3 "Modulation Meter" to FM.

70.10.4 "Dev Kc Switch" to 150.

70.10.5 "Int. Mod" to FM.

70.10.6 "Frequency cps" to 30 Kc.

70.10.7 Adjust "FM Modulation Control" until the "Modulation Meter" reads 40 Kc on the 150 Kc scale.

70.11 Connect the receiver output to the Panoramic Telemetering Indicator input jack.

70.12 Make the following adjustments to the Panoramic Telemetering Indicator.

70.12.1 "Sweep Range Selector" to "Log."

70.12.2 "Scale Selector" to "Lin."

70.12.3 "Scale Multiplier" to "X 10."

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70.12.4 Adjust the "Input Level Controls" until the 40 Kc signal is full scale as read on the Panoramic Telemetering Indicator.

70.13 Disconnect the Boonton Signal Generator output from the receiver input and reconnect the receiver input to the antenna.

70.14 On the Boonton turn the "DBM Microvolt Control" to 0.

70.15 Turn S2 on and connect the DC VTVM between the positive and negative terminals of the 240-280 VDC power supply. Insure the DC VTVM indicates +260 volts. Adjust 240-280 volt power supply if necessary.

70.16 Insure that the receiver is tuned to the transmitter operating frequency.

70.17 Adjust R136 to obtain a full scale reading of the 40 Kc signal as indicated on the Panoramic Telemetering Indicator.

Full scale equals 40 Kc $\pm$ 4 Kc.

(18) 40 Kc $\pm$ 4 Kc

70.18 Adjust R132 until the amplitude of the 3.9 Kc signal indicated on the Panoramic Telemetering Indicator is 8 Kc $\pm$ 1 Kc.

(19) 8 Kc $\pm$ 1 Kc

70.19 On the Panoramic Telemetering Indicator set the "scale multiplier" to "X 1".

70.20 Adjust R134 until the amplitude of the 1.2 Kc signal indicated on the Panoramic Telemetering Indicator is 2 Kc $\pm$ 0.5 Kc.

80.0 INTERMODULATION DISTORTION

80.1 Make the following adjustments to the Panoramic Telemetering Indicator.

80.1.1 "Scale Selector" to "Log."

80.1.2 Adjust the "input level controls" until the 40 Kc signal is full scale as read on the Panoramic Telemetering Indicator.

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80.2 Verify that all harmonic distortion and intermodulation products are at least 34 db below full scale amplitude of the 40 Kc signal indicated on the Panoramic Telemetering Indicator.

(20)_____
Initial

80.3 Turn S1 and S2 OFF and remove all test equipment loads from the OA-1208(__)/DPN Modulator Assembly.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-23369B(AS) TRANSMITTER GROUP, TELEMETRIC DATA OA-1208,( )/DPN

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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